Artificial Intelligence for Smart-Building Log Analysis

PROJECT PROPOSAL

Group Number: C262T-4201 | Ticket Number: D01 | Host Organisation: Smart Facilities SA

Mentors: Wenhao Liang and Abhilash Sridhara | Week 1 starts: 03-Aug-2026

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Date: 27 August 2026

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EXECUTIVE SUMMARY

Smart Facilities SA is the host organisation for this project. The project will investigate the use of artificial intelligence and machine-learning techniques for analysing smart-building operational data. The publicly available CU-BEMS Energy Research Data from the Chamchuri 5 building at Chulalongkorn University will be used as the primary research dataset. The dataset contains minute-level measurements collected across seven building floors and multiple zones, including air-conditioning, lighting and plug-load power consumption, together with indoor temperature, relative humidity and illuminance measurements. The selected dataset is used as a representative development and evaluation dataset for the project prototype and does not represent the full range of smart-building systems described in the Smart Facilities SA project environment.

The project will develop and evaluate a reproducible data-analysis pipeline for integrating and analysing these heterogeneous measurements. The pipeline will support data preparation, data integration, feature extraction, pattern analysis, anomaly detection, visualisation and evaluation. Python will be used as the

main programming language, together with open-source data science and machine-learning libraries. The project will compare interpretable statistical baseline methods with several suitable unsupervised anomaly-detection approaches. This will allow the final recommendations to be based on evidence from different methods rather than relying on a single model.

The analysis will examine operational patterns at building, floor, zone and device levels where supported by the available data. The project will investigate potentially unusual patterns such as unexpected changes in equipment power consumption, unusual temporal behaviour and deviations from typical relationships between energy consumption and environmental conditions. Because the selected dataset does not provide comprehensive verified records of equipment faults or abnormal incidents, detected anomalies will not automatically be treated as confirmed real-world failures. Evaluation will instead use controlled synthetic anomalies, stability and consistency testing, alert-volume analysis and qualitative review of representative results.

The main outputs will include a documented data-integration and analysis pipeline, reusable source code, a canonical analysis-ready dataset or reproducible transformation process, exploratory analysis results, anomaly-detection results and visualisations. The project will also develop a framework for interpreting and prioritising detected anomalies based on factors such as magnitude, persistence, recurrence, location and supporting context. A final technical report will summarise the results, limitations and recommendations for possible future operational use.

The project is planned over ten teaching weeks. No mandatory software-licensing cost is expected because the main tools are open source. An optional budget of AUD 200 is included for cloud computing if local or organisational computing resources are not sufficient. Total student effort is estimated at about 400 hours. This is recorded separately as an estimated labour-value item rather than a direct project cost.

1. PROJECT BACKGROUND AND DESCRIPTION

Smart Facilities SA is the host organisation for this project. Its facilities-management environment includes a range of smart-building systems, such as access-control devices, heating and cooling systems, environmental sensors, lifts, power-management systems and building-management servers. The project focuses on the use of artificial intelligence and machine-learning techniques to support anomaly detection in smart-building operations.

To provide a realistic and reproducible basis for the research, the project will use publicly available smart-building operational data collected through the CU-BEMS building energy management system at the Chamchuri 5 building of Chulalongkorn University. The selected dataset will serve as a representative development and evaluation dataset for the project prototype and does not represent the full range of smart-building systems described in the broader Smart Facilities SA project environment.

The selected dataset contains minute-level measurements collected from a seven-storey academic office building between July 2018 and December 2019. The available data includes power consumption from air-conditioning, lighting and plug-load systems, together with indoor environmental measurements such as temperature, relative humidity and illuminance. Measurements are available across multiple floors and zones, although the number and type of available devices and sensors vary between locations (Pipattanasomporn et al., 2020).

Analysing this type of smart-building data is not straightforward. Different floors and zones have different equipment configurations and operating conditions, while normal behaviour can vary according to time, location, device type and environmental conditions. The data also contains a large volume of minute-level observations that would be difficult to inspect manually. For these reasons, fixed rules or manual inspection may not consistently identify unusual operational behaviour. Previous research has examined data-driven anomaly detection in building and energy environments, where large volumes of sensor and operational data can be used to identify patterns and potential abnormal behaviour (Himeur et al., 2021).

This project will examine whether artificial intelligence and machine-learning methods can distinguish normal operational behaviour from unusual patterns in the selected building energy and environmental data. Potential anomalies may include unexpected changes in equipment power consumption, unusual temporal patterns, deviations from typical environmental conditions, or inconsistent relationships between related variables. The project will not assume that one detection method is suitable for all cases. Instead, interpretable statistical baselines and selected machine-learning approaches will be developed and compared using the selected dataset. This is necessary because anomaly-detection methods use different assumptions about normal and abnormal behaviour, and their performance can vary between datasets and application domains (Chandola et al., 2009).

The project will also consider how detected anomalies can be presented with sufficient context to support interpretation and prioritisation in a future operational environment. Because the selected dataset does not provide comprehensive verified records of equipment faults, security incidents or other abnormal events, a detected anomaly will not automatically be interpreted as a confirmed real-world incident. Instead, evaluation will use controlled synthetic anomalies, consistency and stability checks, alert-volume analysis and qualitative review of representative cases.

The project objectives are to:

- Create a reproducible workflow for importing, integrating, cleaning, transforming and analysing smart-building data across multiple floors and zones.
- Develop a consistent data structure for integrating heterogeneous measurements, including equipment power consumption and environmental sensor data.
- Identify normal temporal, device-level, zone-level and cross-variable operational patterns.
- Develop and compare suitable anomaly-detection methods, including statistical baselines and selected machine-learning approaches.
- Create visualisations that show operational patterns, anomaly timing, affected floors, zones or systems, and supporting evidence.
- Evaluate the detection methods using controlled synthetic anomalies, consistency and stability checks, alert-volume analysis and qualitative review where appropriate.
- Develop an interpretation and prioritisation approach that could support facilities or security staff in reviewing and triaging detected anomalies in a future operational environment.

The expected benefits for Smart Facilities SA include improved visibility of smart-building operational patterns and a structured approach for identifying potentially unusual behaviour. The project may help demonstrate how anomaly-detection methods could support the prioritisation of investigation effort by facilities or security staff in a future operational environment. The results will also provide evidence about the usefulness, limitations and practical requirements of AI-assisted monitoring, including the requirements for extending the prototype to other smart-building or security-related data sources before any larger-scale production system is considered.

2. PROJECT SCOPE

2.1 Scope Rationale

- The project scope focuses on the main tasks required for analysing smart-building operational and sensor data and is designed to be achievable within the available ten-week project timeframe. It covers data integration and preparation, feature engineering, exploratory pattern analysis, anomaly detection, evaluation, visualisation and interpretation. These areas are included because the performance of anomaly-detection methods depends on the quality, structure and characteristics of the available data (Chandola et al., 2009).

- The project will use the CU-BEMS Energy Research Data from the Chamchuri 5 building as its primary research dataset. The analysis will focus on the available measurements across multiple floors and zones, including air-conditioning, lighting and plug-load power consumption, indoor temperature, relative humidity and illuminance. The selected dataset represents a subset of the broader smart-building environment described in the Smart Facilities SA project context. Therefore, the prototype will focus on the available CU-BEMS measurements while considering how the approach could be extended to other operational or security-related data sources in a future deployment.

- The project will compare interpretable statistical baseline methods with selected machine-learning approaches. It will not assume that one method is suitable for every variable, device or operating condition. Visualisation and anomaly prioritisation are also included so that detected patterns can be interpreted in their building and operational context and could support facilities or security staff in reviewing representative findings in a future operational environment.

- Production deployment, automated building control, and cybersecurity testing are outside the scope of this project. These exclusions keep the project focused on developing and evaluating an analytical prototype rather than creating a live operational system.

2.2 Inclusions

- The following aspects of the project are considered within scope:

- Review and characterisation of the selected CU-BEMS dataset, including available floors and zones, measurement variables, timestamps, sampling frequency, missingness, data volume and quality constraints.
- Data cleaning and preparation, including timestamp processing, duplicate handling, missing-value treatment, data-type standardisation and integration of measurements from different floors, zones and systems into an analysis-ready structure.
- Feature engineering at appropriate time, floor, zone and device levels, such as rolling statistics, moving averages, rate of change, temporal features, power-consumption patterns and relationships between energy consumption and environmental measurements.
- Exploratory analysis and pattern discovery to identify normal operating patterns, daily and weekly cycles, recurring behaviour, differences between floors or zones, and relationships between building energy consumption and environmental conditions.
- Development of interpretable statistical baseline methods and one or more suitable machine-learning anomaly-detection approaches for predominantly unlabeled smart-building operational data.
- Comparison and evaluation of candidate methods using controlled synthetic anomalies, stability and consistency testing, alert-volume analysis and qualitative review of representative results.
- Visualisation of normal operating patterns and detected anomalies, with sufficient contextual information to show their timing, location, affected variables and supporting evidence.
- Development of an interpretation and prioritisation framework based on available evidence, such as anomaly magnitude, persistence, recurrence, affected floor or zone and relationships with other measurements, to support potential review and triage by facilities or security staff in a future operational environment.
- Documentation, source-code packaging and final presentation of findings, limitations and recommendations.

2.3 Exclusions

- The following aspects of the potential project are explicitly out of scope unless later agreed through formal scope control:

- Deployment of a production, 24/7 monitoring platform within live building-management infrastructure.
- Direct control of physical building equipment or automated execution of operational responses.
- Replacement of existing building-management, energy-management, maintenance or monitoring systems.
- Procurement or installation of new sensors, controllers, networking equipment or other building hardware.
- Long-term collection of new operational data beyond the selected research dataset.
- Formal cybersecurity penetration testing, vulnerability exploitation or forensic investigation.

- Direct validation of security incidents, access-control events or lift-system anomalies, as these data types are not available in the selected CU-BEMS dataset. Their inclusion would require additional approved data sources and further development.
- Confirming that every detected anomaly represents an actual equipment fault or operational failure, as the selected dataset does not provide comprehensive verified incident labels.
- Guaranteeing prediction of all future equipment failures or security incidents; the project is an analytical investigation and prototype evaluation.

3. RISKS

The most significant project risks arise from the quality and structure of the selected smart-building dataset, the absence of verified anomaly labels, the complexity of integrating measurements across floors and zones, and the limited ten-week project timeframe. These risks may affect both model validity and project delivery. The project will therefore conduct an early data audit, maintain interpretable baseline methods, document uncertainty explicitly and avoid dependence on any single modelling technique.

Risk	Probability (out of 1.0)	Impact	Mitigation Strategy
Missing, inconsistent or noisy measurements in the selected dataset	0.70	High; may add 20–30 hours and reduce model reliability	Perform a data-quality audit early in the project; quantify missingness; identify invalid or inconsistent values; define cleaning and imputation rules; preserve the original data and document all transformations.
Few or no trustworthy labels for true anomalies or equipment faults	0.80	High; limits conventional accuracy measurement	Use unsupervised methods; compare against statistical baselines; inject controlled synthetic anomalies where appropriate; conduct stability and consistency testing; analyse representative results and explicitly report evaluation limitations.
Complexity in integrating data across multiple floors, zones and measurement types	0.65	Medium–High; may add 20–30 hours	Develop a canonical analysis structure; standardise timestamps and variable names; document mappings between floors, zones and measurements; test individual subsets before full integration.
Large data volume or computational limitations caused by minute-level measurements	0.50	Medium–High; may slow experimentation and model evaluation	Begin with selected floors, zones or time periods during development; optimise data types and processing steps; aggregate data where justified; use optional cloud computing only if existing resources are insufficient.
Synthetic anomalies may not accurately represent real equipment faults	0.60	High; may limit the practical interpretation of evaluation results	Use multiple types and magnitudes of controlled anomalies; base scenarios on plausible changes in power consumption or environmental measurements; distinguish clearly between simulated anomalies and confirmed real-world faults.
Excessive false positives make anomaly results difficult to interpret	0.65	High; reduces practical value	Compare alert volumes; tune thresholds; consider persistence and recurrence; rank anomalies rather than treating all detections equally; provide contextual visualisations.
Model results are unstable across different floors, zones or time periods	0.45	Medium–High; may weaken conclusions	Use time-based validation and comparisons across different data subsets; test multiple algorithms and parameter settings; conduct sensitivity analysis; prefer simpler and more stable methods when performance is similar.
Scope expansion or uneven team workload	0.40	Medium; could delay milestones	Maintain a defined project backlog and weekly milestone review; assign clear workstream owners; prioritise core deliverables; discuss major scope changes with the mentor before implementation.

4. BUDGET

The project is designed to minimise direct expenditure by using existing student/organisational computing resources and open-source software. Actual and potential project cash expenditure and hypothetical labour-value estimates are separated below. Any optional expenditure will require client/mentor approval before commitment.

Budget Item	Estimated Cost	Justification
Public CU-BEMS research dataset	AUD 0 (actual)	The selected smart-building dataset is publicly available for research and analysis, so no data acquisition or licensing cost is expected.
Python/R and open-source analytics libraries	AUD 0 (actual)	No licence fee is expected for the core programming, data preparation, visualisation and machine-learning toolchain.
Existing student/university/host computing resources	AUD 0 incremental (actual)	The project will use available laptops, university facilities or approved host resources where practical.
Secure project storage using approved existing services	AUD 0 incremental (actual)	Use organisation/university-approved storage already available to the project; no new storage purchase is assumed.
Optional cloud compute contingency	Up to AUD 200 (potential actual)	Only if local or approved existing resources are insufficient for selected experiments; expenditure is not assumed unless approved.
Student project effort: 5 members x 8 h/week x 10 weeks = 400 h	AUD 16,000 (hypothetical)	Illustrative labour value at AUD 40/hour. This is not an expected cash cost to Smart Facilities SA.
Presentation / final handover materials	AUD 0-50 (potential actual)	Electronic delivery is preferred; small printing or presentation costs may arise only if required.
Estimated mandatory project cash cost	AUD 0	The planned project can be completed using open-source software and existing resources, subject to data-access and computing arrangements.

5. ROLES AND RESPONSIBILITIES

The allocation below reflects the tasks initially completed by each team member during Week 1 and establishes the primary workstreams for the remainder of the project. Responsibilities are distributed according to the main technical and project requirements, including data preparation and integration, feature engineering, anomaly detection, technical integration, evaluation, project coordination, documentation and presentation.

Each team member has a defined area of responsibility while remaining jointly responsible for collaboration, integration, peer review and the overall quality of the final project outputs. Responsibilities may be refined as the characteristics of the selected dataset, technical requirements and stakeholder feedback become clearer.

Team Member	Primary Role / Focus	Responsibilities
Huaye Qi (Qi) a1948935	Project Leader & Machine Learning Engineer	• Coordinate project activities, team progress and major task dependencies. • Lead key technical decisions and communication with mentors/client. • Research, select and implement suitable anomaly-detection methods. • Compare statistical baselines and machine-learning approaches. • Guide the integration and communication of major technical findings.
Yongli Jiang (Lee) a1906489	Data Engineer	• Analyse the structure and quality of the selected smart-building operational and sensor data. • Develop the data structure and data-preparation process for integrating measurements across floors, zones and systems.

Team Member	Primary Role / Focus	Responsibilities
		• Perform data cleaning, transformation and timestamp processing. • Develop relevant features for pattern analysis and anomaly detection. • Support exploratory analysis of normal operational patterns.
Jiahao Wang (Gavin) a1940040	AI Systems Engineer & Evaluation Lead	• Integrate data preparation, feature engineering and anomaly-detection components into the analysis pipeline. • Design the system workflow and experimental evaluation approach. • Evaluate model outputs, false-positive burden, stability and limitations. • Compare methods and support evidence-based model selection. • Interpret overall technical results and recommendations.
Meng Zhang (Penelope) a1947217	Project Coordination & Presentation Support	• Organise project materials and meeting outputs. • Prepare presentation slides, diagrams and visual summaries based on team technical outputs. • Support the communication of project progress and findings. • Assist with preparation of the final presentation materials.
Zirui Geng (Roc) a1941715	Documentation & Project Support Officer	• Maintain project documentation, timeline tracking and milestone records. • Organise shared project materials and supporting evidence. • Assist with report formatting, references and documentation consistency. • Support final submission and handover preparation.
Wenhao Liang and Abhilash Sridhara	Project Mentors	Provide academic and methodological guidance, review project progress and assumptions, and advise the team on scope, risks and project direction.
Smart Facilities SA	Client / Host Organisation	Provide the operational context and application perspective for the project; clarify requirements and review the usefulness of project outputs.

5.1 Individual Contributions and Learning Plan

The team will monitor individual contributions through assigned workstreams, weekly progress reviews, shared documentation and milestone-based deliverables. Each team member will maintain responsibility for their primary work area while contributing to integration, peer review and the final project outputs. The project is also intended to support individual learning through the practical application of technical and project-management knowledge.

Team Member	Planned Contribution	Expected Learning
Huaye Qi (Qi) a1948935	Lead project coordination and technical direction, while researching, selecting and supporting the implementation of suitable anomaly-detection methods.	Develop practical knowledge of technical project leadership, statistical and machine-learning approaches to anomaly detection, model selection and communication of technical findings.
Yongli Jiang (Lee) a1906489	Analyse the selected smart-building operational and sensor data and support data profiling, cleaning, transformation, integration and feature engineering for the analysis pipeline.	Develop practical skills in understanding heterogeneous operational data, identifying data-quality issues, data preparation and feature engineering.
Jiahao Wang (Gavin) a1940040	Integrate the main technical components of the analysis pipeline and support the evaluation and comparison of anomaly-detection methods and results.	Develop practical skills in AI system integration, experimental evaluation, robustness analysis and evidence-based interpretation of analytical results.
Meng Zhang (Penelope) a1947217	Organise project materials and meeting outputs and support the preparation of presentation slides, diagrams and visual summaries based on team technical outputs.	Develop practical skills in project coordination, presentation preparation and communicating project progress and technical findings.
Zirui Geng (Roc) a1941715	Support the preparation of visual materials, technical documentation, project tracking and final report, presentation and handover materials.	Develop practical skills in project coordination, technical documentation, presentation preparation and structured communication of project results.
Wenhao Liang and Abhilash Sridhara	Project Mentors	Provide academic and methodological guidance, review project progress and assumptions, and advise the team on scope, risks and project direction.
Smart Facilities SA	Client / Host Organisation	Provide operational context and an application perspective, clarify project requirements where necessary, and review the practical relevance and usefulness of project outputs and recommendations.

Progress and learning will be reviewed at regular team meetings and project milestones. Where a team member encounters difficulties, delays, or critical technical blockers (such as data integration issues or model convergence challenges), an explicit escalation path is triggered: such risks will be flagged within 24 hours to coordinate ad-hoc reviews, ensuring milestones are actively managed without compromising assigned workstream ownership. Individual contributions will be evidenced through project artefacts such as research notes, data-analysis outputs, code contributions, experimental results, visualisations, documentation, report sections and presentation materials.

5.2 Communications Plan

The following communication plan has been established for the project and will be reviewed and adjusted as needed throughout the project period.

Stakeholder / Group	Purpose	Frequency	Method
Student team	Coordinate tasks, dependencies, code integration and risks	Weekly 45-60 minute meeting; asynchronous updates as needed	In-person/online meeting plus agreed group chat; shared repository and task board
Smart Facilities SA client representative	Confirm project requirements, application context, operational relevance and key project decisions.	Weekly or fortnightly 30-minute checkpoint, with additional questions consolidated asynchronously	Online/in-person meeting and email; concise agenda and action log
Project mentor	Review methodology, scope, risks and academic progress	Weekly or fortnightly checkpoint in line with course arrangements	Scheduled meeting and email; milestone artefacts shared before review
Facilities/security subject-matter staff, if nominated	Provide operational context and, where available, assess the potential relevance and interpretability of representative anomaly outputs.	Targeted sessions during Weeks 2, 7 and 9 or as agreed	Interview/workshop; structured examples and visualisations
Whole stakeholder group	Confirm milestone completion and final acceptance	Milestone reviews and final Week 10 presentation	Presentation/demo plus final report and handover package

5.3 Project Stakeholders

Name	Role	Interest in the project	Responsibilities
Smart Facilities SA client representative	Client / Sponsor	Receives project outputs; concerned with operational value, feasibility, risk, privacy, interpretability and whether the prototype supports future decision-making.	Confirm project requirements, provide operational direction and review key project outputs and recommendations.
Facilities-management staff	Primary operational users	Need useful identification of abnormal building/equipment behaviour without excessive false positives or unexplained model outputs.	Provide operational context for interpreting normal and potentially abnormal building behaviour, where possible, and review the practical usefulness of representative anomaly outputs.
Security staff	Potential operational users	Interested in whether identified anomalies and unusual building behaviour could provide useful context for future security investigation and prioritisation.	Provide a security perspective on how anomalies could be interpreted, prioritised and potentially investigated in a future operational environment.
Maintenance / engineering staff	Subject-matter users	Interested in early indicators of equipment degradation, recurring faults and abnormal operating patterns that may support maintenance planning.	Provide equipment and maintenance context and assess whether representative anomaly patterns could potentially indicate equipment faults, degradation or abnormal operating conditions in a future operational environment.
IT / data administrators	Data and system custodians	Concerned with secure data handling, access permissions, log schemas, system performance and reproducibility of data extraction/processing.	Advise on potential data-handling, access, integration and system requirements that would need to be considered if the prototype were extended to a live organisational environment.
Wenhao Liang and Abhilash Sridhara	Academic stakeholder	Interested in appropriate methodology, project governance, evidence-based evaluation and achievement of project objectives.	Provide academic supervision, methodological guidance and feedback on project progress and deliverables.

Name	Role	Interest in the project	Responsibilities
Student project team	Delivery team	Responsible for completing the ten-week project, integrating technical workstreams and delivering a coherent, defensible outcome.	Complete the agreed workstreams, integrate project outputs, manage risks and deliver the final prototype, documentation and presentation.

6. DELIVERABLES AND PROJECT EVALUATION CRITERIA

The project will deliver the following items electronically to Smart Facilities SA using the publicly available CU-BEMS Energy Research Data as the primary research dataset.

Deliverable	Description	Evaluation Criteria
D1. Data preparation and feature-engineering pipeline	Reusable Python scripts/notebooks that load and integrate the selected CU-BEMS data, perform cleaning, timestamp processing and data standardisation, and generate documented analysis features.	Runs reproducibly on the selected CU-BEMS dataset; key transformations and assumptions are documented; data-quality issues are quantified.
D2. Exploratory pattern analysis	Analysis of temporal behaviour, energy-consumption patterns, environmental conditions, floor and zone differences, and other recurring operational patterns in the selected building data	Identifies understandable normal patterns and provides evidence that supports subsequent anomaly modelling.
D3. Anomaly-detection model comparison	A comparison of at least one interpretable statistical baseline and one or more suitable machine-learning anomaly-detection methods, including documented configurations, anomaly outputs and comparative analysis.	Methods are evaluated consistently; false-positive burden, stability and limitations are reported; selected method(s) are justified by evidence.
D4. Visualisation package	Dashboard-style views of the selected CU-BEMS data will provide an overall building summary, including total or average power consumption, average indoor temperature, relative humidity, illuminance, and detected-anomaly counts. Energy views will show air-conditioning, lighting and plug-load consumption over time, with comparisons across floors and zones. Environmental views will show temperature, relative humidity and illuminance over time and across floors and zones to provide contextual evidence. Anomaly monitoring will display the timestamp, affected floor and zone, variable or system, anomaly magnitude or score, duration or persistence, recurrence, and relevant energy or environmental context. Filters will allow users to select the time period, floor, zone, and measurement type or system.	A reviewer can filter by time period, floor, zone and measurement type or system, locate and interpret representative detected anomalies without reading raw logs, and trace each view to the underlying CU-BEMS measurements. Visual outputs are legible and do not present detected anomalies as confirmed faults.
D5. Interpretation and prioritisation framework	A practical structure for explaining detected anomalies and ranking investigation priority based on available evidence such as magnitude, persistence, recurrence, affected floor or zone, subsystem context and any available anomaly score.	Potential facilities or security reviewers can distinguish higher-priority from lower-priority findings and understand why an anomaly was surfaced.
D6. Final technical report and recommendations	Report covering data, methodology, experiments, results, limitations, evaluation approach and recommendations for potential future application in a live smart-building environment.	Conclusions are traceable to analysis results; limitations are explicit; recommendations distinguish prototype findings from production requirements.
D7. Reproducible handover package and presentation	Source code, environment/dependency information, README/run instructions, selected outputs and final presentation/demo.	Another technically capable reviewer can understand the project structure and reproduce the agreed analysis using the same CU-BEMS dataset and documented environment.

Project success will be judged primarily by reproducibility, technical validity, stability of the anomaly-detection methods, interpretability of outputs, usefulness of the prioritisation framework and the quality of evidence supporting conclusions.

Although the chosen CU-BEMS dataset focuses on comprehensive energy and environmental benchmark records, in the absence of verified on-site equipment fault labels, conventional classification metrics (such as accuracy, recall, and F1-score) will serve only as auxiliary references rather than definitive standards for actual deployment performance. Consequently, the evaluation framework will adopt a multi-faceted approach, including controlled baseline consistency checks, anomaly stability analysis, alert frequency distribution evaluations, cross-method comparisons, and qualitative reviews of representative cases by facilities management experts, ensuring the business relevance and interpretability of the prototype system

7. IMPLEMENTATION PLAN

1. Dataset and project requirements confirmation (Week 1, Milestone M1): Confirm project objectives, stakeholder expectations, selected CU-BEMS dataset scope, success criteria and technical environment. Establish the repository, task board and reproducibility conventions.

2. Data audit and analysis structure (Weeks 1–2, M2): Profile the available CU-BEMS files, floors, zones, variables, timestamps, sampling frequency, data volume, missingness and device configurations. Define the data structure and mappings required for analysis across selected data sources.

3. Data preparation pipeline (Weeks 2–3, M3): Implement data loading, cleaning, timestamp processing, duplicate handling, missing-value treatment, standardisation and validation checks. Produce an analysis-ready dataset or reproducible transformation process.

4. Feature engineering and exploratory pattern discovery (Weeks 3–4, M4): Create temporal, rolling-statistical, rate-of-change and cross-variable features. Examine daily and weekly cycles, correlations, differences between floors and zones, recurring operating patterns and relationships between energy consumption and environmental measurements.

5. Baseline anomaly detection (Weeks 4–5, M5): Implement interpretable statistical baselines such as thresholds, z-scores, robust scores or rolling-deviation approaches to establish reference behaviour and identify potential anomalies.

6. Machine-learning anomaly modelling (Weeks 5–6, M6): Implement and compare suitable unsupervised anomaly-detection methods such as Isolation Forest (Liu et al., 2008), Local Outlier Factor (Breunig et al., 2000) and other justified techniques. More complex methods will only be included if the data volume and structure support their use.

7. Evaluation, tuning and robustness analysis (Weeks 6–7, M7): Apply controlled synthetic anomalies, time-based validation where appropriate, consistency and stability checks and comparison between representative methods. Tune thresholds and model parameters with attention to alert volume and interpretability.

8. Visualisation and anomaly prioritisation (Weeks 7-8, M8): Develop dashboard-style views of air-conditioning, lighting and plug-load power consumption; indoor temperature; relative humidity; illuminance; and detected anomalies. Support filtering or comparison by time period, floor, zone and measurement type or system. For representative anomalies, show timing, affected floor or zone, affected variable or system, magnitude or available anomaly score, duration or persistence,

recurrence and relevant contextual measurements. Use this evidence to support interpretation and prioritisation.

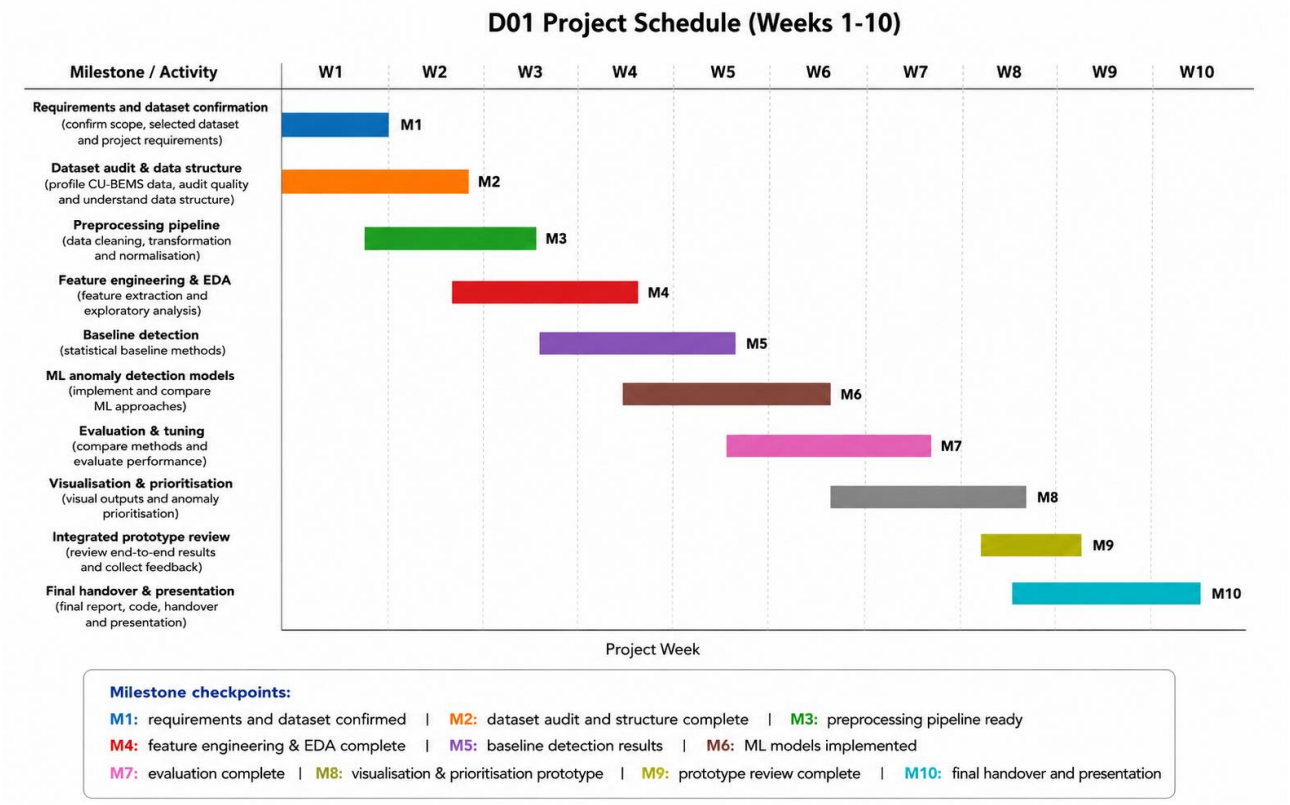
9. Integrated prototype review (Week 9, M9): Run the end-to-end pipeline on the selected dataset, review representative outputs, assess technical and usability limitations, and correct high-priority issues before finalisation.

10. Finalisation and handover (Week 10, M10): Finalise reproducible code, complete documentation and the technical report, prepare the final presentation or demonstration, document limitations and provide recommendations for potential future deployment in a live smart-building environment.

11. The schedule assumes that the selected CU-BEMS dataset remains suitable for the planned analysis after the initial data audit. If significant data-quality issues, missing measurements or structural limitations are identified, the team will adjust the selected floors, zones, variables, feature set or analytical methods while maintaining the overall project objective of developing and evaluating a smart-building anomaly-analysis prototype.

8. PROJECT SCHEDULE

The schedule below runs from Monday of Week 1 through Friday of Week 10, as required by the template. Milestones correspond to the implementation plan above.



9. REFERENCES

Breunig, M. M., Kriegel, H.-P., Ng, R. T., & Sander, J. (2000). LOF: Identifying density-based local outliers. In *Proceedings of the 2000 ACM SIGMOD international conference on management of data* (pp. 93–104). Association for Computing Machinery. https://doi.org/10.1145/342009.335388

Chandola, V., Banerjee, A., & Kumar, V. (2009). Anomaly detection: A survey. *ACM Computing Surveys, 41*(3), Article 15, 1–58. https://doi.org/10.1145/1541880.1541882

Himeur, Y., Ghanem, K., Alsalemi, A., Bensaali, F., & Amira, A. (2021). Artificial intelligence based anomaly detection of energy consumption in buildings: A review, current trends and new perspectives. *Applied Energy, 287*, 116601. https://doi.org/10.1016/j.apenergy.2021.116601

Liu, F. T., Ting, K. M., & Zhou, Z.-H. (2008). Isolation forest. In *2008 eighth IEEE international conference on data mining* (pp. 413–422). IEEE. https://doi.org/10.1109/ICDM.2008.17

Pipattanasomporn, M., Chitalia, G., Songsiri, J., Aswakul, C., Pora, W., Suwankawin, S., Audomvongseree, K., & Hooncharoen, N. (2020). CU-BEMS, smart building electricity consumption and indoor environmental sensor datasets. *Scientific Data, 7*, Article 241. https://doi.org/10.1038/s41597-020-00582-3

APPROVAL

We approve the project as described above.

Name	Role (e.g. student, client, mentor)	Signature	Date
Huaye Qi (Qi)	Student	Huaye Qi	27/08/2026
Yongli Jiang (Lee)	Student	Yongli Jiang	27/08/2026
Jiahao Wang (Gavin)	Student	Jiahao Wang	27/08/2026
Meng Zhang (Penelope)	Student	Meng Zhang	27/08/2026
Zirui Geng (Roc)	Student	Zirui Geng	26/08/2026
[Client representative name]	Client		
Wenhao Liang and Abhilash Sridhara	Mentor		